Introduction to Photogrammetry

Lecture 05

Introduction to Photographic Cameras

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In Today’s Class

1. Introduction
2. Characteristics of Photogrammetric Cameras
3. Types of Photogrammetric Cameras

1. Introduction

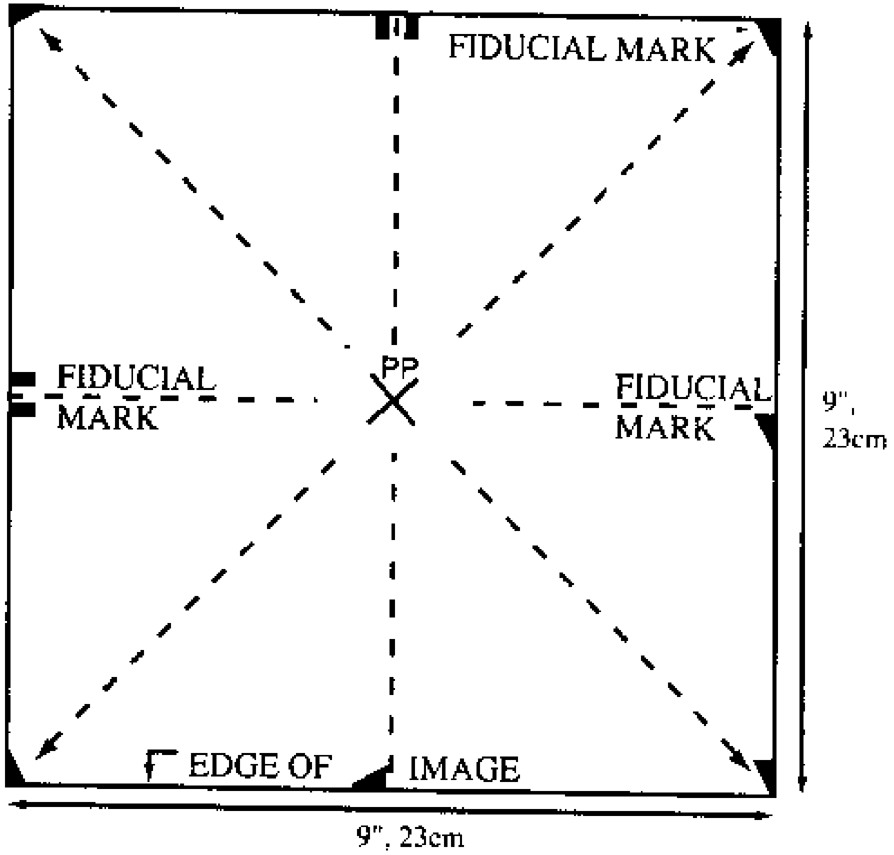
- Cameras are the primary instruments used to capture images for photogrammetry.
- These cameras record photographs that later serve as data for measurement and analysis.
- The accuracy of photogrammetric measurements strongly depends on the quality and design of the camera.
- Therefore, special cameras are developed specifically for photogrammetric purposes.

- Before digital sensors became common, most aerial photographs were captured using **film-based cameras**.
- These cameras used photographic film to record images.
- Film cameras played an important role in early aerial mapping and surveying.
- Even today, understanding film-based cameras helps students understand the **fundamental principles of photogrammetry**.

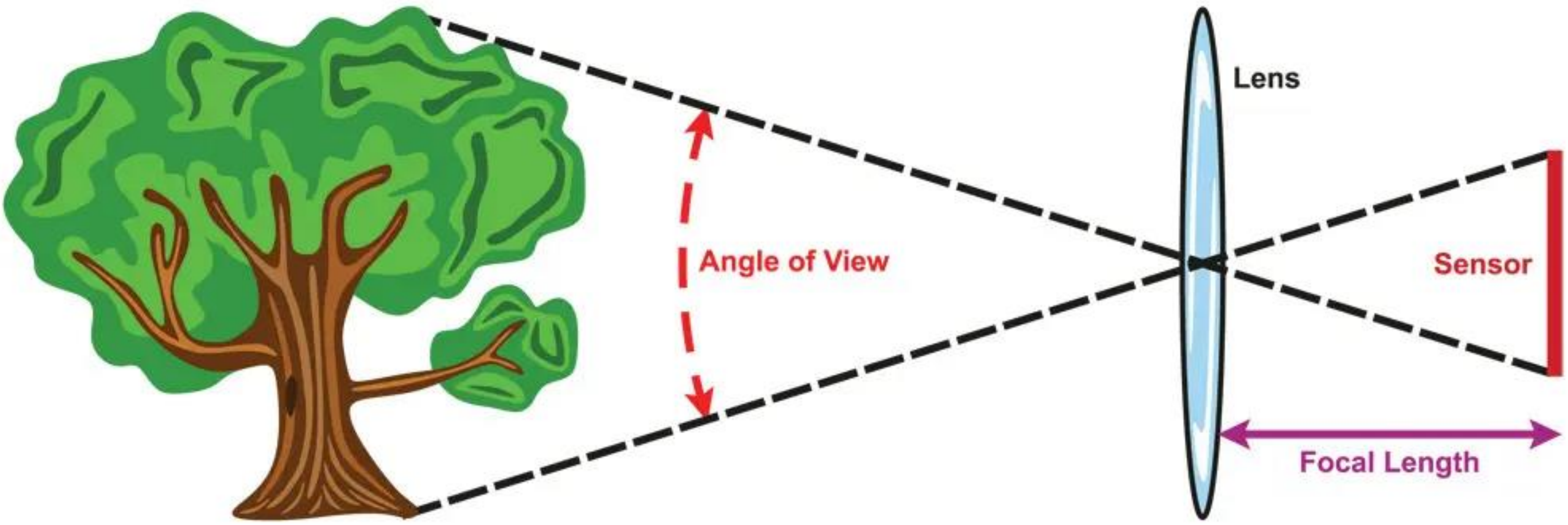
- Photogrammetric cameras are specially designed cameras used for accurate mapping and measurements.
- These cameras are also known as **metric cameras**.
- They maintain precise geometric relationships between the lens and the image plane.
- This stability allows reliable measurements to be made from the photographs.

2. Characteristics of Photogrammetric Cameras

- Photogrammetric cameras have several important characteristics.
- They maintain **stable internal geometry**.
- They have a **precisely known focal length**.
- They contain **reference marks called fiducial marks**.
- These features make the photographs suitable for accurate measurements.

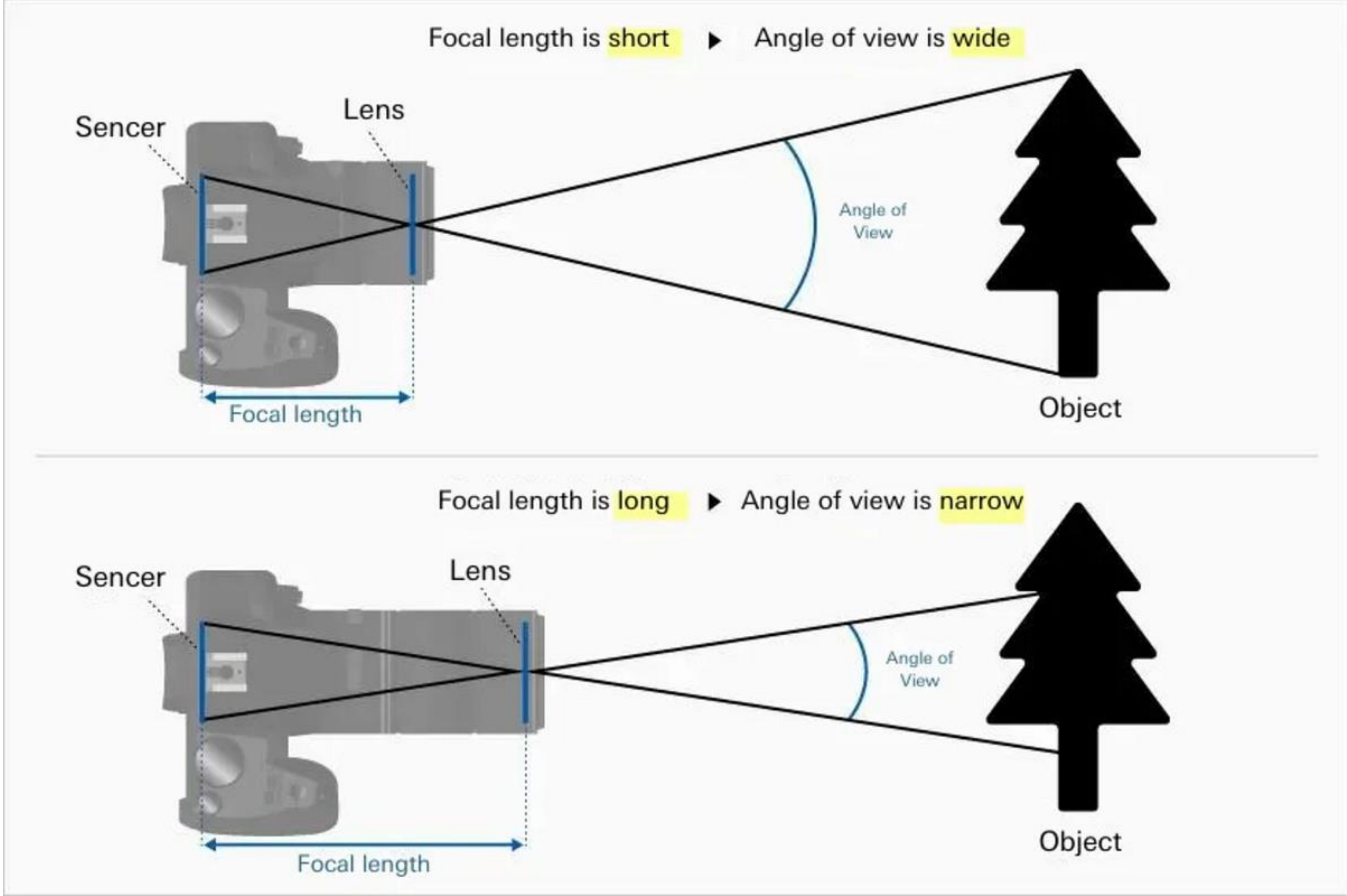


Focal Length and Angle of View



The greater the focal length, the closer the image and the less the angle of view

The less the focal length, the farther the image and the greater the angle of view


Focal length is short ▶ Angle of view is wide
Sencer
Lens
Angle of View
Object
Focal length
Focal length is long ▶ Angle of view is narrow
Sencer
Lens
Angle of View
Object
Focal length

3. Types of Photogrammetric Cameras

- Photogrammetric cameras can be classified based on their location of use.
- The two main categories are **aerial cameras** and **terrestrial cameras**.
- Aerial cameras capture images from aircraft.
- Terrestrial cameras capture images from ground-based positions.
- In mapping applications, aerial cameras are used most frequently.

Aerial Cameras

- Aerial cameras are mounted on aircraft or drones to capture images of the Earth’s surface.
- These cameras are specifically designed for **mapping and surveying purposes**.
- Because of their mapping function, they are sometimes called **cartographic cameras**.
- Aerial photographs obtained from these cameras are widely used for producing maps.

Applications of Aerial Photography

- Aerial photographs have many practical applications.
- They are used for **topographic map production**.
- They help monitor **urban growth and land use change**.
- They assist in **agriculture and forestry management**.
- They support **environmental monitoring and disaster management**.

Metric Cameras

- Metric cameras are cameras designed for accurate photogrammetric measurements.
- These cameras have carefully controlled internal geometry.
- The position of the lens relative to the film plane remains fixed.
- This stability ensures accurate measurement of image coordinates.

Non-Metric Cameras

- Non-metric cameras are ordinary cameras that are not designed for precise measurements.
- Examples include handheld cameras and most consumer cameras.
- These cameras may introduce distortions that make accurate measurements difficult.
- However, modern processing techniques sometimes allow limited photogrammetric use of non-metric cameras.

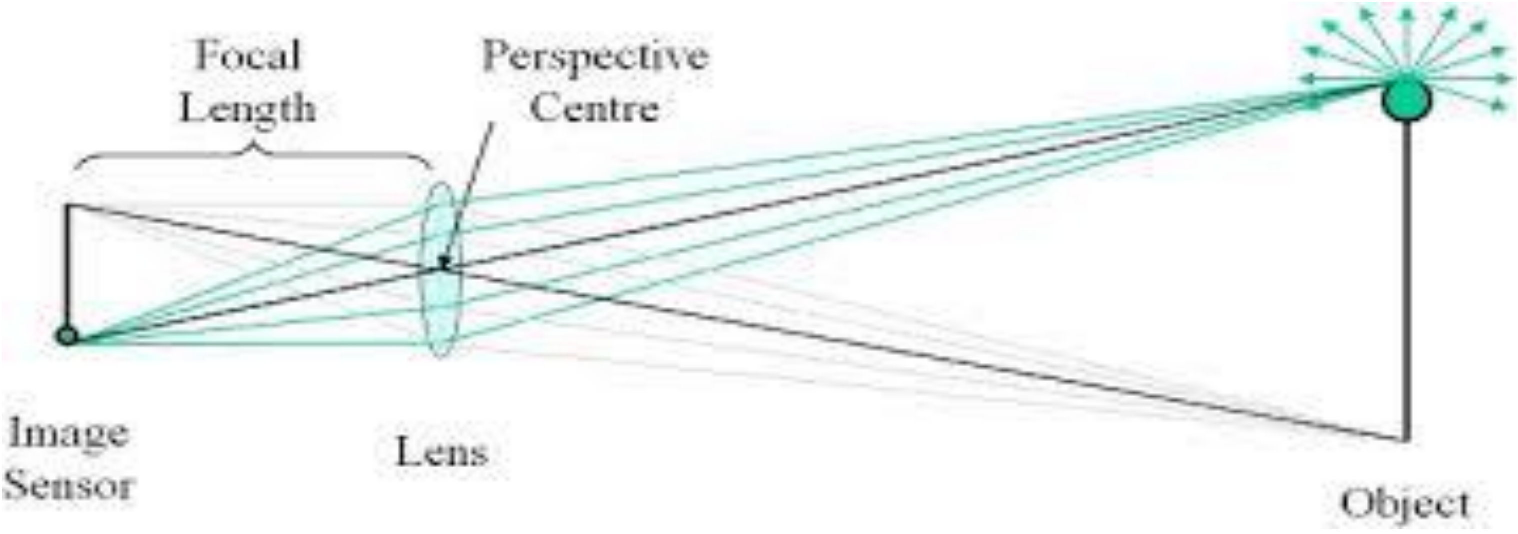
Film-Based Photography

- Film-based cameras use photographic film to record images.
- The film is coated with light-sensitive chemicals that react when exposed to light.
- After exposure, the film is developed chemically to produce the photograph.
- These photographs can then be analyzed for photogrammetric measurements.

- Aerial cameras typically use large film formats.
- A common format is **23 cm × 23 cm (9 inch × 9 inch)**.
- Larger film size allows more ground area to be captured in a single photograph.
- Large images also improve measurement accuracy.

Basic Principle of Image Formation

- Photographic cameras work based on the principle of **central perspective projection**.
- Light rays from objects pass through the camera lens.
- These rays converge at a point called the **perspective center**.
- The image is then formed on the film plane.



Perspective Geometry

- In perspective geometry, objects farther from the camera appear smaller.
- Objects closer to the camera appear larger.
- This effect is known as **perspective distortion**.
- Photogrammetry uses mathematical methods to correct these distortions.

Ground Coverage of Aerial Photographs

- The area covered by an aerial photograph depends on several factors.
- These include the **flying height of the aircraft**.
- They also depend on the **focal length of the camera lens**.
- Higher altitude results in larger ground coverage.

Scale of Aerial Photographs

- The scale of an aerial photograph represents the relationship between distances on the photograph and distances on the ground.
- Photo scale depends on the **flying height and focal length**.
- Larger scale images show more detail.
- Smaller scale images cover larger areas but with less detail.

Importance of Camera Stability

- Photogrammetric cameras must remain stable during exposure.
- Any movement of the camera may blur the image.
- Stability ensures sharp and accurate photographs.
- Special mounting systems are used to stabilize aerial cameras.

Thank you!

Any Questions